Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

EARTH TESTER · HAND-DRIVEN · METAL BODY

CIE/222M

3-terminal and 4-terminal, any range 0–1 Ω to 0–10,000 Ω

Earth resistance and soil resistivity, in the field

Direct measurement of the resistance of earth in power circuits, telecommunications, railway electrification, domestic and industrial electrical installations, lightning protection installations and various other electrical systems. The tester performs equally well where soil conductivity is low, for example in mountainous areas, and where powerful stray earth currents exist. The four-terminal instrument is designed for straightforward, approximate soil investigation in agricultural and geological research and in planning large civil engineering projects such as tunnels, drainage, sewerage schemes and tube railways.



CIE/222M · ILLUSTRATIVE IMAGE

3 / 4	±5%	1–5
TERMINAL TYPES	ACCURACY	RANGES PER INSTRUMENT

01 HIGHLIGHTS

- ✓ Silent generator with collapsible handle
- ✓ Chuck nut locking arrangement prevents anti-clockwise rotation
- ✓ Anti-magnetic aluminium body, protected metallic flap over the scale window
- ✓ Accuracy conforms to IS 9223:1979
- ✓ Available in 3-terminal and 4-terminal types
- ✓ Centrifugal clutch (governor) for constant output voltage
- ✓ Wide anti-parallax mirror scale
- ✓ Accurate, reliable and quick response
- ✓ Long life and strength against vibration and impact
- ✓ Single range or multi-range with range selector switch

02 TYPICAL APPLICATIONS

- Earth resistance measurement for power circuits and lightning protection systems
- Telecommunications and railway electrification earthing checks
- Domestic and industrial electrical installation earthing
- 4-terminal type: soil resistivity surveys for civil engineering projects (tunnels, drainage, sewerage)

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03 HOW IT WORKS

PRINCIPLE

Test current comes from a smooth, noiseless hand-driven A.C. generator. A high-grade centrifugal governor holds the voltage constant, so variation in crank speed does not affect the reading.

CONSTRUCTION

Housed in a sturdy, elegant anti-magnetic cast-aluminium case with a metallic carrying handle. The instrument is dust-free, rubber shoes give it stability, and a protected metallic flap covers the glass window of the scale. The generator handle folds back flush into the casing.

04 SALIENT FEATURES

Immune to stray currents

Measuring accuracy is not affected by stray alternating current or by the back e.m.f. of earth electrolysis.

Centrifugal governor

Holds the output voltage constant regardless of crank speed.

3-terminal or 4-terminal

Three terminals for earth resistance, four terminals for earth resistance plus specific soil resistivity.

Other resistance work

Can also measure non-inductive and non-capacitive resistance.

05 TECHNICAL SPECIFICATIONS

Body	Anti-magnetic aluminium
Generator Type	Hand-driven generator with centrifugal governor
Terminal Types	3-terminal (earth resistance) or 4-terminal (earth resistance + soil resistivity)
Scale Length	80 mm (approx.)
Accuracy	±5% (as per IS: 9223/1979)
Generator Rotation Speed	160 r.p.m.
Weight	2.5 kg (approx.)
Dimensions (L×W×H)	110 × 128 × 190 mm (approx.)
High Voltage Test	2 KV A.C. (r.m.s.) for 1 min. between electrical circuit and containing case
Insulation Resistance Test	Better than 25 MΩ at 500 V D.C. between electrical circuit and containing case

Ranges are listed on the following pages. Specifications are approximate where marked.

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06 SINGLE RANGE

RANGE BAND	SELECTION
0–1 Ω to 0–5 Ω	Any range within this band
0–10 Ω to 0–100 Ω	Any range within this band
0–200 Ω to 0–10,000 Ω	Any range within this band

Any single range from 0–1 Ω up to 0–10,000 Ω can be specified within the three bands above.

07 DUAL RANGE

CONFIGURATION	FULL-SCALE RANGES (Ω)
Dual range	0–1 / 10 · 0–2 / 10 · 0–3 / 30 · 0–10 / 100

08 3-TERMINAL AND 4-TERMINAL

3-terminal type

Measures earth resistance. Connect the earth electrode under test and two test spikes. Test kit: 2 M.S. spikes (45 cm), 3 flexible-wire spools (10, 20 and 30 m), hammer and carrying bag, at extra cost.

4-terminal type

Measures earth resistance and specific soil resistivity, using four spikes in a line. Test kit: 4 M.S. spikes (45 cm), 4 flexible-wire spools (10, 20, 20 and 30 m), hammer and carrying bag, at extra cost.

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09 TRIPLE, FOUR AND FIVE RANGE

CONFIGURATION	FULL-SCALE RANGES (Ω)
Triple range	0–1 / 10 / 100 · 0–2 / 10 / 100 · 0–2 / 20 / 200 · 0–3 / 30 / 300 · 0–10 / 100 / 1,000
Four range	0–1 / 10 / 100 / 1,000 · 0–2 / 10 / 100 / 1,000 · 0–10 / 100 / 1,000 / 10,000
Five range	0–1 / 10 / 100 / 1,000 / 10,000 · 0–2 / 10 / 100 / 1,000 / 10,000

Multi-range instruments carry a range selector switch, to be chosen from 0–1 Ω to 10,000 Ω. Any other range can be manufactured against specific requirements.

10 OPTIONAL ACCESSORIES (AT EXTRA COST)

- Testing kit, 3-terminal: 2 M.S. spikes (45 cm), 3 flexible-wire spools (10 / 20 / 30 m), hammer, carrying bag
- Testing kit, 4-terminal: 4 M.S. spikes (45 cm), 4 flexible-wire spools (10 / 20 / 20 / 30 m), hammer, carrying bag
- Carrying case for earth tester

11 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/222M, 4-terminal, 0–1 / 10 / 100 Ω**. Any range not listed can be manufactured against specific requirements.

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